

# Graph Theory – Lecture Notes

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Summer Semester 2026

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# 1 Introduction

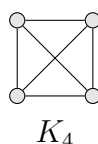
**Definition 1.1.** A (*simple*) *graph*  $G$  is a pair  $(V, E)$  where  $V$  is a non-empty set and  $E \subseteq \{\{x, y\} \in V \times V : x \neq y\}$ . We call the elements of  $V$  *vertices* of  $G$  and the elements of  $E$  *edges* of  $G$ . We also write  $V(G)$  for  $V$  and  $E(G)$  for  $E$ , respectively.

‘Simple’ means no loops and no double edges.

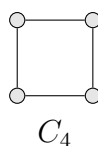
**Remark 1.2.**  $V$  is finite, unless otherwise stated.

**Example 1.3.** We define a few common graphs. Let  $n, m \in \mathbb{N}$ .

- The *complete graph*  $K_n$  has  $n$  vertices and each pair of vertices is connected.



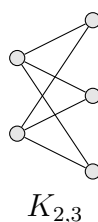
- The *cycle*  $C_n$  has  $n$  vertices, which are cyclically connected.



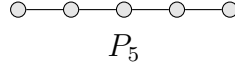
- Let  $A$  and  $B$  be disjoint sets with  $|A| = n$  and  $|B| = m$ . The *complete bipartite graph*  $K_{m,n}$  is defined by

$$V = A \dot{\cup} B \tag{1}$$

$$E = \{\{a, b\} : a \in A, b \in B\} \tag{2}$$



- The *path*  $P_n$  has the vertices  $1, \dots, n$  and edges  $\{a, a+1\}$  for  $a \in \{1, \dots, n-1\}$ . (in books sometimes  $P_{n-1}$  has  $n$  vertices)



**Definition 1.4.** Let  $G$  be a graph,  $u, v \in V(G)$

- If  $\{u, v\} \in E(G)$ , then  $u$  and  $v$  are **adjacent** and the edge  $e = \{u, v\}$  is **incident** with both  $u$  and  $v$ .
- $N_G(v) = \{w \in V(G) : \{v, w\} \in E(G)\}$  is the **neighbourhood** of  $v$  in  $G$ .
- $\deg_G(v) = |N_G(v)|$  is the **degree** of  $v$ .

**Lemma 1.5** (Handshake Lemma). *Let  $G$  be a graph. Then*

$$\sum_{v \in V(G)} \deg_G(v) = 2|E(G)|. \quad (3)$$

(in particular the sum of the degree is even).

*Proof.*  $\sum_{v \in V(G)} \deg_G(v) = \#\{(v, e) : v \in V(G), e \in E(G) \text{ such that } v \text{ incident to } e\} = 2|E(G)|$  □

**Definition 1.6.** Two graphs  $G = (V, E)$  and  $G' = (V', E')$  are **isomorphic** if there exists an isomorphism  $\phi$  between them, i.e. a bijection  $\phi : V \rightarrow V'$  such that  $\forall u, v \in V(G) : \{u, v\} \in E(G) \iff \{\phi(u), \phi(v)\} \in E(G')$

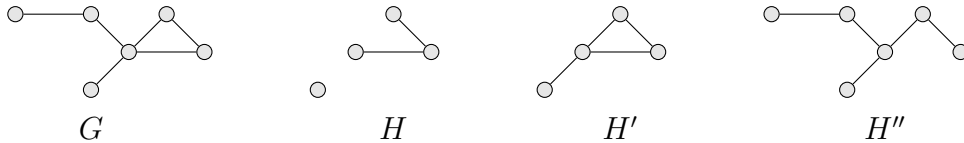
**Definition 1.7.** Let  $G = (V, E)$  be a graph. A **subgraph** of  $G$  is a graph  $H = (V', E')$  with  $V' \subseteq V$  and  $E' \subseteq E \cap \binom{V'}{2}$  where  $\binom{S}{K}$  is the set of all  $K$ -subsets of  $S$ .

$H'$  is an **induced subgraph** of  $G$  if it is a subgraph and  $E' = E \cap \binom{V'}{2}$

$H''$  is a **spanning subgraph** if it is a subgraph and  $V' = V$

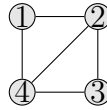
For  $W \subseteq V$ , write  $G[W]$  for the subgraph induced by the vertices in  $W$

**Example 1.8.**



**Definition 1.9.** Let  $v_i$  be vertices of a graph  $G$

- A **walk** is a sequence of vertices  $v_i$  s.t.  $\{v_i, v_{i+1}\} \in E(G) \forall i \in \{1, \dots, k-1\}$
- The **length** of a walk is  $k$  (number of vertices in the walk)
- The walk is **closed** if  $v_1 = v_k$
- A **trail** is a walk that does not use an edge twice
- A **path** is a walk for which  $v_1, \dots, v_k$  are pairwise distinct



$(1, 2, 3, 4, 2, 3)$  is a walk, but no trail

$(1, 2, 3, 4, 2)$  is a trail, no path

$(1, 2, 3, 4)$  is a path

**Definition 1.10.** Let  $G$  be a graph

- A graph  $G$  is **connected** if there is a walk between any pair of its vertices
- A **(connected) component** of  $G$  is an inclusion wise maximal connected subgraph

## 1.1 Hamiltonian Cycles

**Definition 1.11.** A **Hamiltonian cycle** in a graph  $G$  is a spanning subgraph that is a cycle (isomorphic to  $C_n$  for some  $n \in \mathbb{N}$ )

If  $G$  has a Hamiltonian cycle,  $G$  is called **Hamiltonian**

**Theorem 1.12** (Dirac's Theorem). *Let  $G$  be a graph of order  $n \geq 3$  where the **order** of  $G$  is given by  $|V(G)|$  and the **size** by  $|E(G)|$ . If every Vertex of  $G$  has degree at least  $\frac{n}{2}$  then  $G$  is Hamiltonian*

**Theorem 1.13** (Ore's theorem). *Let  $G$  be a graph of order  $n \geq 3$ . If for every pair  $u, v \in V, u \neq v, u$  and  $v$  non-adjacent, we have  $\deg(u) + \deg(v) \geq n$  then  $G$  is Hamiltonian*

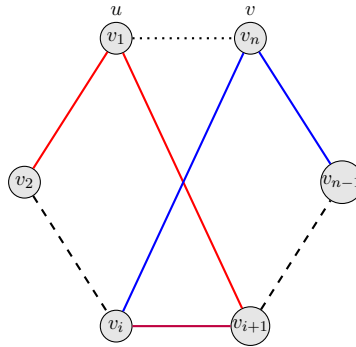
**Remark 1.14.**Ore  $\implies$  Dirac*Proof.*

The Graph  $G$  is connected: suppose  $G$  is not connected. Take an arbitrary component of  $G$  and call it  $H$ . The order of  $H$  is  $n - k$  if  $k$  vertices are not in  $V(H)$ . Now any  $v \in V(H)$  has  $\deg(v) \leq n - k - 1$  with equality if  $v$  is adjacent to every vertex in  $V(H)$ . Now take  $u \notin V(H)$  so  $\deg(u) \leq k - 1$ . It follows that  $u, v \in V(G), u \neq v$  and  $u, v$  are not adjacent so  $\deg(u) + \deg(v) \geq n$  but  $\deg(u) + \deg(v) \leq k - 1 + n - k - 1 = n - 2 < n$   $\nmid$

- $G$  has a Hamiltonian cycle: By induction on the number of non-edges
- No non-edge  $\implies G$  complete, hence Hamiltonian
- Assume  $G$  non complete  $u, v \in V, u \neq v$ , be non-adjacent. By induction the graph  $G'$  obtained from  $G$  by adding  $e = \{u, v\}$  as an edge is Hamiltonian. Let  $C$  be a Hamiltonian cycle of  $G'$ . If  $C$  does not use  $e \implies C$  is Hamiltonian cycle in  $G$ . Assume  $C$  uses  $e$ . Removing  $e$  from  $C$  yields a path from  $u$  to  $v$

For  $i \in \{2, \dots, n - 1\}$ 

- If  $v_i$  is a neighbour of  $u$ , color  $\{v_{i-1}, v_i\}$  red
- If  $v_i$  is a neighbour of  $v$ , color  $\{v_i, v_{i+1}\}$  blue
- Since  $\deg(u) + \deg(v) \geq n$  and the path  $v_1, \dots, v_n$  has  $n - 1$  edges, there exists an edge  $\{v_i, v_{i+1}\}$  which is both red and blue. Then  $v_1, \dots, v_i, v_n, v_{n-1}, \dots, v_{i+1}$  is a Hamiltonian cycle in  $G$ .



□

## 1.2 Eulerian Trail

**Definition 1.15.** An **Eulerian trail** is a trail that passes through each edge of the graph exactly once. A graph with a closed Eulerian trail is called **Eulerian**

**Theorem 1.16** (Euler's theorem). *Let  $G$  be a connected graph with at least 2 vertices. Then  $G$  has a closed Eulerian trail iff every vertex in  $G$  has even degree*

*Proof.*

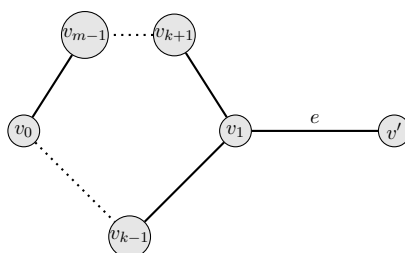
"  $\implies$  "

Let  $T$  be a closed Eulerian trail and fix an orientation on  $T$ . Then for every  $v \in V(G)$ , the incident edges can be classified as "ingoing" or "outgoing". This defines a bijection between in and outgoing edges of  $T$ , so  $\deg(v)$  is even.

"  $\impliedby$  "

Let  $T = (v_0, e_1, v_1, \dots, e_m, v_m)$  be a longest trail. We will prove (i)  $v_0 = v_m$  and (ii)  $\{e_i : i \in \{1, \dots, m\}\} = E(G)$ . (i)+(ii)  $\iff$   $T$  is closed Eulerian trail.

- (i) If  $v_0 \neq v_m$ , then  $v_0$  is incident to an odd number of edges in  $T$ . As  $\deg(v_0)$  is even, there is  $e \in E(G)$  incident to  $v_0$  and not on  $T$ . We could thus append  $e$  on  $T$   $\nmid$  because  $T$  is longest trail.
- (ii) Claim:  $V(T) = V$ . Assume otherwise there exists a vertex  $v' \notin V(T)$  and  $e = \{v_k, v'\} \in E(G)$  for some  $k \in \{0, \dots, m\}$  (by connectedness). Then  $(v', e, v_k, e_{k+1}, v_{k+1}, \dots, v_{m-1}, e_m, v_0, e_1, \dots, v_{k-1}, e_k)$  is a longer trail hence  $V(T) = V$



Claim:  $E(T) = E$ . Assume otherwise. Consider  $e \in E \setminus E(T)$  and write  $e = \{v_k, v_l\}$ . Then  $(v_k, e_{k+1}, v_{k+1}, \dots, v_{m-1}, e_m, v_0, e_1, v_1, \dots, e_k, v_k, e, v_l)$  is a longer trail  $\nmid$

□

**Theorem 1.17.** *Let  $G$  be a connected graph with at least two vertices. Then  $G$  has an Eulerian trail iff there are precisely 0 or two vertices of odd degree*

*Proof.*

By the Euler's Theorem, we know that if there are 0 vertices of odd degree then there exists an Eulerian trail (in fact a closed Eulerian trail), and if there exists a closed Eulerian trail, then all vertices have even degree.

Suppose now that  $G$  has exactly two vertices  $x$  and  $y$  of odd degree. We consider the following graph  $G'$ : Set  $V(G') = V(G) \cup \{z\}$ , where  $z \notin V(G)$ . Furthermore set  $E(G') = E(G) \cup \{\{x, z\}, \{y, z\}\}$ . Clearly, every vertex has even degree in  $G'$ , so there exists a closed Eulerian trail. Since  $z$  is only adjacent to  $x$  and  $y$ , these are the neighbours of  $z$  on the trail. Deleting the edges  $\{x, z\}$  and  $\{y, z\}$  from the trail yields an Eulerian trail from  $x$  to  $y$  in  $G$ . Conversely, if there is a non-closed Eulerian trail with endpoints  $x$  and  $y$  in  $G$ , then  $x$  and  $y$  are non-adjacent. Adding the edge  $\{x, y\}$  creates a graph with a closed Eulerian trail, so all vertices have even degree by Euler's theorem. Thus  $x$  and  $y$  are the only vertices in  $G$  with odd degree.  $\square$

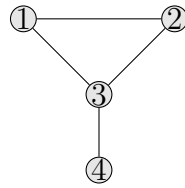
### 1.3 Datastructures

Assume that  $G = (V, E)$  is a graph with  $v = \{1, \dots, n\}$

**Definition 1.18.** The **adjacency matrix** of  $G$  is the matrix  $A = (a_{ij})$  with 
$$a_{ij} = \begin{cases} 1 & \{i, j\} \in E \\ 0 & \text{else} \end{cases}$$

The **adjacency list** of  $G$  stores a label for each vertex, and a list of (pointers to) its neighbours.

**Example 1.19.**



$$A = \begin{pmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \end{pmatrix}$$

- $v_1$ :  $v_2, v_4$



- $v_2$ :  $v_1, v_3, v_4$
- $v_3$ :  $v_2$
- $v_4$ :  $v_1, v_2$

**Remark 1.20.**

Memory requirement:

Adjacency matrix:  $\theta(n^2)$

Adjacency list:  $\theta(n + m)$ , where  $m$  is the number of edges

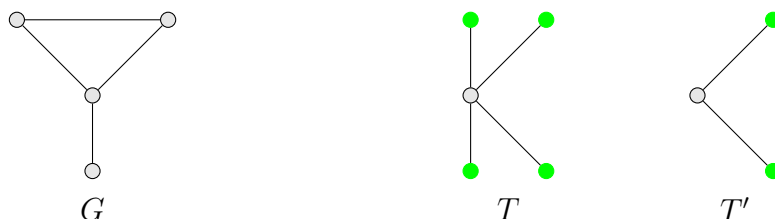
## 2 Trees

**Definition 2.1.** A **tree** is a graph that is connected and does not contain a cycle (a graph "contains a cycle" if it has a subgraph isomorphic to  $C_n$  for some  $n \geq 3$ ).

A **forest** is a graph that does not contain a cycle.

A **leaf** is a vertex of a tree with degree 1.

**Example 2.2.**



- $G$  is not a tree nor a forest
- $T, T'$  are trees and forests
- $T \cup T'$  is a forest but not a tree
- All green vertices are leafs, all white vertices are not leafs

**Lemma 2.3.** *Every tree  $T$  with  $n \geq 2$  vertices has at least one leaf.*

*Proof.* We iteratively construct a path  $v_1, \dots, v_n$  as follows:

Start with any edge  $\{v_1, v_2\} \in E(T)$ . Assume that we have constructed  $v_1, \dots, v_k$ . To construct  $v_{k+1}$ , we may assume that  $v_k$  is not a leaf in  $T$  (or we are done). Otherwise  $v_k$  has a neighbour  $v_{k+1} \neq v_{k-1}$ .  $v_{k+1} \neq v_i$  because  $T$  is a tree and otherwise we would get a cycle  $(v_i, \dots, v_k, v_{k+1} = v_k)$ . Add  $v_{k+1}$  to the path and continue finite many steps (because there are only finitely many vertices). The algorithm just stated ends at a leaf.  $\square$

**Theorem 2.4.** *Let  $G$  be a graph with  $n$  vertices. The following are equal:*

- (i)  $G$  is a tree
- (ii)  $G$  is connected with  $n - 1$  edges
- (iii)  $G$  is connected and removing an edge disconnects it
- (iv)  $G$  has exactly one path between each pair of distinct vertices

(v)  $G$  has no cycle, but adding an edge creates a cycle

*Proof.* "1.  $\implies$  2."

Induction on  $n$ :

$n = 1$  is trivial

Let  $n \geq 2$ . By Lemma (2.2),  $G$  has a leaf  $v$ . Removing  $v$  and the unique edge incident to  $v$  yields a tree  $G'$ . By induction  $G'$  has  $n - 2$  edges so  $G$  has  $n - 1$  edges.

"2.  $\implies$  3."

Suppose there is an edge  $e \in E(G)$  and the resulting graph by removing  $e$  is still connected. Keep removing as long as the resulting graph is still connected. Let  $G'$  be the resulting graph. Observe that  $G'$  is a spanning subgraph of  $G$ .  $G'$  does not contain a cycle as otherwise we could remove one edge from the cycle. By "1.  $\implies$  2.", applied to  $G'$ , we know that  $G'$  has  $n - 1$  edges  $\nless$  because  $G$  had  $n - 1$  edges and we removed at least one.

"3.  $\implies$  4."

Suppose that there is a pair with  $u, v \in V(G)$  s.t. there are two distinct paths  $u_i, v_i$  between them. Let  $i_0$  be the maximum index s.t.  $u_i = v_i : \forall i \in \{1, \dots, i_0\}$ . The edge  $\{v_{i_0}, v_{i_0+1}\}$  does not lie on the path  $u_i$ . The graph obtained by removing  $\{v_{i_0}, v_{i_0+1}\}$  is connected as for any walk using this edge we can use  $u_i$  to go to  $u$  to reach  $v_{i_0}$  or  $v$  to reach  $v_{i_0+1}$ .

"4.  $\implies$  5."

If  $G$  had a cycle, then there would exist two paths between each pair of neighbouring vertices on the cycle (clockwise + counter clockwise).

Adding an edge creates a cycle: Let  $u, v \in V(G)$  non adjacent. There is one path between the two. Now adding an edge between  $u, v$  creates a second path  $\nless$

"5.  $\implies$  1."

We only need to show that  $G$  is connected. Suppose  $u, v$  are not connected by a walk. Creating an edge between  $u, v$  does not result in a cycle because for being isomorphic to  $C_n$  one needs at least two walks between each of the vertices in the cycle but by assumption the only walk connecting  $u, v$  is the newly added edge  $\nless$   $\square$

**Corollary 2.5.** *Every connected graph  $G$  has a spanning tree.*

*Proof.*

Consider graphs on  $n$  vertices, now do induction on the number of edges  $m$ . As  $G$

is connected,  $m \geq n - 1$ . If  $m = n - 1 \implies G$  is a tree. If  $m > n - 1 \implies G$  is not a tree. By (2.3) we can remove edges s.t. the resulting graph is connected. By induction it has a spanning tree, which is then also a spanning tree of  $G$ .  $\square$

**Theorem 2.6** (Matrix-tree theorem). *Let  $G$  be a graph with vertices  $\{1, \dots, n\}$ , edges  $\{e_1, \dots, e_m\}$ .*

*Let  $Q = (q_{ij})$  with  $q_{ij} = \begin{cases} -1 & \{i, j\} \in E \\ 0 & i \neq j : \{i, j\} \notin E \\ \deg(i) & i = j \end{cases}$  Let  $Q_{ij}$  be the matrix obtained from  $Q$  by removing the  $i$ -th row and the  $j$ -th column. Then the number of labelled spanning trees of  $G$  is  $|\det(Q_{ij})|$ , for any choice of  $i, j$ .*

**Corollary 2.7** (Cayley's formula). *The number of labelled spanning trees of  $K_n$  is  $n^{n-2}$  if  $n \geq 2$ .*

*Proof.* Define  $Q$  as in (2.5).

$$Q_{11} = \begin{pmatrix} n-1 & 1 & \cdots & 1 \\ 1 & \ddots & & \vdots \\ \vdots & & \ddots & 1 \\ 1 & \cdots & 1 & n-1 \end{pmatrix} \rightarrow \bar{Q}_{11} = \begin{pmatrix} n-1 & -1 & -1 & \cdots & -1 \\ -n & n & 0 & \cdots & 0 \\ -n & 0 & \ddots & & \vdots \\ \vdots & \vdots & & \ddots & 0 \\ -n & 0 & \cdots & 0 & n \end{pmatrix} \rightarrow \hat{Q}_{11} =$$

$$\begin{pmatrix} 1 & \star & \cdots & \star \\ 0 & n & & \vdots \\ \vdots & & \ddots & \star \\ 0 & \cdots & 0 & n \end{pmatrix}$$

$$\implies |\det(Q_{11})| = |\det(\hat{Q}_{11})| = n^{n-1-1} = n^{n-2} \quad \square$$

### 3 Planar Graphs

**Definition 3.1.** A **planar drawing** of a graph  $G$  consists of injective continuous functions

$$\phi_v : V(G) \rightarrow \mathbb{R}^2 \qquad \phi_e : [0, 1] \rightarrow \mathbb{R}^2$$

s.t.:

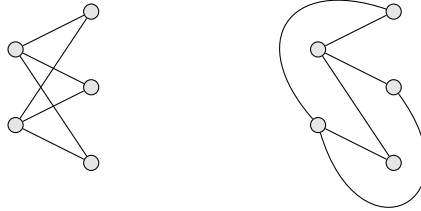
- $\{\phi_{\{u,v\}}(0), \phi_{\{u,v\}}(1)\} = \{\phi_u(u), \phi_v(v)\}$  for all  $\{u, v\} \in E(G)$
- The sets  $\phi_e((0, 1))$  for each  $e \in E(G)$  are pairwise disjoint and disjoint from  $\text{Im}(\phi_v)$  (i.e. no crossings)

A graph  $G$  is called **planar** if it has a planar drawing.

A **plane graph** is a planar graph, together with a fixed planar drawing.

**Example 3.2.**

- Trees
- $K_4$
- $K_{2,3}$



**Definition 3.3.** A **face** of a plane graph  $G$  is a maximal connected part of  $\mathbb{R}^2 \setminus (\text{Im}(\phi_v(u)) \cup \text{Im}(\phi_e))$ . Every plane graph has exactly one unbounded face called the **outer face**. All other faces are called **inner faces**. The **boundary** of a face are the vertices and edges of  $G$  incident with it. A plane graph is called **planar triangulation** if every face is bounded by a cycle of length precisely 3 (this includes the outer face).

**Theorem 3.4** (Euler's formula). *Let  $G = (V, E)$  be a connected planar graph. Then the number of faces  $|F|$  of  $G$  is  $|F| = |E| - |V| + 2$ .*

*Proof.* Induction on  $|E|$ .

Since  $G$  is connected, we have  $|E| \geq |V| - 1$  so we start the induction at  $|E| = |V| - 1 \implies G$  is a tree. Since a tree does not contain a cycle, we have  $|F| = |E| - |V| + 2$ . Now let  $|E| > |V| - 1$ . Then  $G$  is not a tree, so  $G$  contains a cycle  $C$ . Choose an edge  $e$  on  $C$  and remove it to obtain a plane graph  $G'$ .  $e$  separates two faces in  $G$ , which fuse to a single face in  $G'$ . By induction, the number of faces  $|F'|$  in  $G'$  is  $|F| - 1 = |F'| = |E| - 1 - |V| + 2 \implies |F| = |E| - |V| + 2$ .  $\square$

**Lemma 3.5.** *For every plane graph  $G = (V, E)$  with  $|V| \geq 3$ , we have  $|E| \leq 3|V| - 6$ .*

*Proof.* Wlog  $G$  is connected. Consider a planar drawing of  $G$  and let  $F$  be the set of faces. Every face is bounded by at least 3 edges and each edge bounds at most 2 faces. By Euler's formula (3.4), we obtain  $\frac{2}{3}|E| \geq |F| = |E| - |V| + 2 \implies |E| \leq 3|V| - 6$ .  $\square$

**Remark 3.6.** This means that the number of edges in a planar graph grows only linear proportional to the number of vertices.

**Example 3.7.**

- $K_5$  is not planar (10 edges) due to (3.4)
- $K_{3,3}$  is not planar (without proof)

**Definition 3.8.** A **subdivision** of a graph  $G$  is a graph obtained from  $G$  by repeatedly subdividing some of its edges.

**Theorem 3.9** (Kuratowski's theorem). *A graph  $G$  is planar iff  $G$  contains no subgraph that is a subdivision of  $K_5$  or  $K_{3,3}$ .*

**Remark 3.10.** A subdivision of a graph is planar iff the graph is planar

**Definition 3.11.** Let  $G$  be a graph. The graph obtained from  $G$  by contracting  $e = \{u, v\} \in E(G)$  denoted  $G/e$ , is the graph obtained from  $G$  by deleting  $u, v$ , inserting a new vertex  $w$  and making  $w$  adjacent to all vertices in  $N_G(u) \cup N_G(v)$ . A graph  $G'$  is a **minor** of  $G$  if  $G'$  can be obtained from  $G$  by a sequence of vertex deletions, edge deletions and edge contractions.

**Remark 3.12.** If  $G'$  is a minor of  $G$  then  $G'$  planar  $\iff G$  is planar.

**Theorem 3.13** (Wagner's theorem). *A graph  $G$  is planar iff  $K_{3,3}$  and  $K_5$  are not a minor of  $G$ .*

## 4 Flows, Cuts, Connectivity

**Definition 4.1.** A **directed graph** is a pair  $(V, E)$  with  $E \subseteq (V \times V) \setminus \{(x, x) : x \in V\}$ . Write  $\delta_G^+(v)$  for edges starting in  $v$  and  $\delta_G^-(v)$  for edges ending in  $v$ .

**Definition 4.2.** A **flow network** is a tuple  $(G, s, t, c)$  where  $G$  is a **directed** graph,  $s, t \in V(G)$ ,  $s \neq t$  and  $c : E \rightarrow \mathbb{R}_0^+$ .

- $s$  is called a **source**
- $t$  is called a **target**
- $\forall e \in E(G) : c(e)$  is the **capacity of  $e$**

A **flow** in a flow network is a function  $f : E \rightarrow \mathbb{R}_0^+$  s.t.

- $f(e) \leq c(e)$  for all  $e \in E(G)$
- $\forall v \notin \{s, t\}, \sum_{e \in \delta^-(v)} f(e) = \sum_{e \in \delta^+(v)} f(e)$  (i.e. what flows in flows out)
- $\forall e \in \delta^-(s) \cup \delta^+(t) : f(e) = 0$  (i.e. nothing flows into a source or out of a target)

The **value** of a flow  $f$  is  $\text{val}(f) = \sum_{e \in \delta^-(v)} f(e) = \sum_{e \in \delta^+(v)} f(e)$

**Remark 4.3.** For  $v \in V$ , write  $f(\delta^+(v)) = \sum_{e \in \delta^+(v)} f(e)$  ;  $f(\delta^-(v)) = \sum_{e \in \delta^-(v)} f(e)$

**Definition 4.4.** A **cut** in a graph/directed graph  $G$  is a partition of  $V(G)$  into two non-empty parts  $A$  and  $B$  (i.e.  $V(G) = A \dot{\cup} B$  and  $A, B \neq \emptyset$ ). For  $s, t \in V(G)$ ,  $s \neq t \implies (A, B)$  is a  $(s, t)$ -cut if  $s \in A, t \in B$ .

If  $(S, T)$  is a  $(s, t)$ -cut, write  $\delta^+(S) = \{e \in E(G) : e \text{ start in } S \text{ ends in } T\} = \delta^-(T)$  and  $\delta^-(S) = \{e \in E(G) : e \text{ starts in } T \text{ and ends in } S\} = \delta^+(T)$ .

The **capacity** of the cut is  $c(S, T) = \sum_{e \in \delta^+(S)} c(e) = c(\delta^+(S)) = c(\delta^-(T))$ .

**Theorem 4.5** (max flow-min cut theorem). *Let  $(G, s, t, c)$  be a flow network. Then  $\max\{\text{val}(f) : f \text{ flow}\} = \min\{c(S, T) : (S, T) \text{ is a } (s, t)\text{-cut}\}$ .*



**Lemma 4.6.** *Let  $f$  be a flow,  $(S, T)$  be a  $(s, t)$ -cut. Then  $\text{val}(f) = f(\delta^+(S)) - f(\delta^-(S))$ .*

*Proof.*

Claim:  $\forall A \subset V : f(\delta^+(A)) - f(\delta^-(A)) = \sum_{v \in A} f(\delta^+(v)) - f(\delta^-(v)) = (*)$

The sum  $f(\delta^+(A)) - f(\delta^-(A))$  does not account for edges  $e \in E(G)$  with start and end in  $A$ . On the right side,  $f(e)$  is counted twice, once with positive and once with negative sign, so the sum vanishes.

Then  $\text{val}(f) = f(\delta^-(t)) \stackrel{(1)}{=} \sum_{v \in T} f(\delta^-(v)) - f(\delta^+(v)) \stackrel{(*)}{=} f(\delta^-(T)) - f(\delta^+(T)) = f(\delta^+(S)) - f(\delta^-(S))$

(1) holds because  $f(\delta^+(t)) = 0$  ( $t$  is a sink) and  $\forall v \notin \{s, t\} : f(\delta^-(v)) - f(\delta^+(v)) = 0$  (by definition).  $\square$

**Corollary 4.7.** *Let  $f$  be a flow,  $(S, T)$  an  $(s, t)$ -cut.*

$\text{val}(f) \leq c(\delta^+(S))$

Thus  $\max\{\text{val}(f) : f \text{ flow}\} \leq \min\{c(S, T) : (S, T) \text{ } (s, t)\text{-cut}\}$ .

4.5.

$\max \text{ flow} \leq \min \text{ cut}$  due to Corollary (4.7)

Let  $f$  maximum flow (i.e. a flow for which  $\text{val}(f)$  is maximal). Let  $c' : E(G) \rightarrow \mathbb{R}_0^+$  where  $c'(e) = c(e) - f(e)$ . Let  $S \subset V(G)$  be defined as the set of reachable vertices<sup>1</sup> according to the following rules:

- $s \in S$
- $x \in V(G)$  is in  $S$  (i.e. reachable) if there is a walk (in the underlying directed graph) from  $s$  to  $x$  traversing  $e_1, \dots, e_k$  s.t.
  - (i)  $c'(e_i) > 0$  ( $\iff f(e_i) < c(e_i)$ ) if  $e_i$  is traversed towards  $x$ .
  - (ii)  $c'(e_i) < c(e_i)$  ( $\iff f(e_i) > 0$ ) if  $e_i$  is traversed towards  $s$ .

If  $S = V(G)$ , we can improve  $f$  as follows:

Let  $e_1, \dots, e_k$  be a walk from  $s$  that makes  $t$  reachable, and  $e_i$  is traversed in orientation  $\sigma_i \in \{\pm 1\}$ , then  $\tilde{f}(e_i) = f(e_i) + \sigma_i \varepsilon$ , for  $i \in \{1, \dots, k\}$  and  $\tilde{f}(e_i) = f(e_i)$  otherwise. This defines a flow  $\tilde{f}$  with  $\text{val}(\tilde{f}) > \text{val}(f)$  for  $\varepsilon > 0$  sufficiently small  $\nmid$  Thus  $t \notin S$ . Set  $T = V \setminus S \neq \emptyset$

- $\forall e \in \delta^+(S) \quad c'(e) = 0$ , i.e.  $f(e) = c(e)$   
 $e = \{x, y\}$  with  $x \in S, y \in T$ . If  $c'(e) \neq 0 \implies e$  satisfies (i)  $\implies y$  reachable  $\nmid$

---

<sup>1</sup>These are the vertices where the flow "is not maximal yet" but has a walk to  $s$ .

- $\forall e \in \delta^-(S) \quad c'(e) = c(e) \iff f(e) = 0$
- $e = \{y, x\}$  with  $x \in S, y \in T$ . If  $c'(e) \neq c(e) \implies e$  satisfies (ii)  $\implies y$  reachable  $\nmid$
- $c(S, T) = \sum_{e \in \delta^+(S)} c(e) = \sum_{e \in \delta^+(S)} f(e) - \sum_{e \in \delta^-(S)} f(e) = f(\delta^+(S)) - f(\delta^-(S)) \stackrel{(4.6)}{=} \text{val}(f) \implies \text{val}(f) \geq c(S, T) \geq \min(s, t)\text{-cut}$

□

## 4.1 Connectivity

**Definition 4.8.** A graph  $G$  is  **$k$ -vertex connected** ( $k \in \mathbb{N}$ ) if it has at least  $k + 1$  vertices and whenever we remove up to  $k - 1$  vertices, the resulting graph is connected.

A graph  $G$  is  **$k$ -edge connected** if whenever we remove up to  $k - 1$  edges, the resulting graph is connected.

The **vertex connectivity (edge connectivity)** is the largest  $k \in \mathbb{N}$  s.t.  $G$  is  $k$ -vertex/edge connected.

**Remark 4.9.**  $G$  graph,  $k$  vertex connectivity,  $\lambda$  edge connectivity,  $\delta$  minimal degree of  $G$ , then  $k \stackrel{(i)}{\leq} \lambda \stackrel{(ii)}{\leq} \delta$

*Proof.* (i) If we have a set of  $\lambda$  edges whose removal makes the res. graph disconnected, then deleting one endpoint for each edge also disconnects the graph.

(ii) Delete all edges incident to a vertex of minimal degree, this disconnects the graph.

□

**Lemma 4.10.**  $G$  directed graph,  $s, t \in V, s \neq t$ . Then the maximal number of edge-disjoint  $(s, t)$ -paths (path which do not have any common edges) is equal to the size of a minimal  $(S, T)$ -cut.

*Proof.* "Max flow-min cut theorem" (4.5) with  $\forall e \in E \quad c(e) = 1$ .

□

**Theorem 4.11** (Menger's theorem for edge-connectivity). Let  $G$  be a graph,  $k \in \mathbb{N}$ .  $G$  is  $k$ -edge-connected iff there are  $k$  edge-disjoint paths between any pair of vertices in  $G$ .

*Proof.*

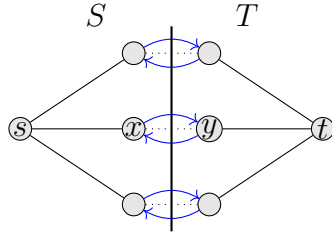
"  $\Leftarrow$  " Removing  $k - 1$  edges will keep one edge intact, so the resulting graph is connected, hence  $G$  is  $k$ -edge connected.

"  $\Rightarrow$  " Let  $s, t \in V, s \neq t$ . Let  $G'$  be the graph obtained from  $G$  by replacing each edge by a double-edge (see graph(1)).

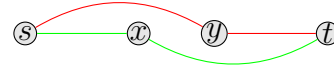
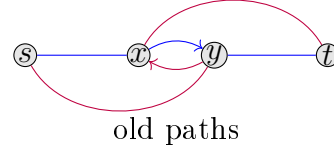
Claim: Any  $(s, t)$ -cut in  $G'$  has at least size  $k$ .

If  $(S, T)$  is an  $(s, t)$ -cut of size  $\leq k - 1$ , then  $G$  has at most  $k - 1$  edges between  $S$  and  $T$ . Removing them disconnects the graph  $\nmid G$  is  $k$ -edge connected. By Lemma (4.10), there are  $k$ -edge disjoint paths  $p_1, \dots, p_k$  from  $s$  to  $t$  in  $G'$ .

Suppose  $x, y \in V$  s.t.  $p_i$  uses  $\{x, y\}$  and  $p_j$  uses  $\{y, x\}$ . Replace  $p_i$  and  $p_j$  by the path following  $p_i$  to  $x$  and continue via  $p_j$  and the path following  $p_j$  to  $y$  and continue via  $p_i$ . Repeat for each such edge. In each step, the sum of the lengths of the paths decreases so the process has to terminate. Then the underlying  $k$ -undirected paths in  $G$  are edge disjoint (see graph(2)).



graph(1)



new paths

graph(2)

□

**Theorem 4.12** (Menger's theorem for vertex-connectivity). *Let  $G$  be a graph with at least  $k + 1$  vertices ( $k \in \mathbb{N}$ ). Then  $G$  is  $k$ -vertex connected iff there are at least  $k$  internally disjoint paths (start and end are the only common vertices) between any pair of distinct vertices of  $G$ .*

*Proof.* "  $\Leftarrow$  " As before (Proof of (4.11)).

"  $\Rightarrow$  " Let  $a, b \in V$  be distinct and  $v_1, \dots, v_n$  be the remaining vertices of  $G$ . Consider the following directed Graph  $G'$  (see graph(1)):

- $V(G') = \{s, t, v'_1, \dots, v'_n, v''_1, \dots, v''_n\}$
- $E(G') = \{\{v'_i, v''_i\} : i \in \{1, \dots, n\}\} \cup \{\{v''_i, v'_j\} : \{v_i, v_j\} \in E(G)\} \cup \{\{v''_j, v'_i\} : \{v_i, v_j\} \in E(G)\} \cup \{\{s, v_i\} : \{a, v_i\} \in E(G)\} \cup \{\{v''_i, t\} : \{b, v_i\} \in E(G)\} \cup \{\{s, t\} \text{ if } \{a, b\} \in E(G)\}$  where we call elements of  $\{\{v'_i, v''_i\} : i \in \{1, \dots, n\}\}$  red edges.

Let  $k_0$  be the minimum size of an  $(s, t)$ -cut in  $G'$ . If  $k_0 \geq k$ ,  $G'$  has  $k$ -edge disjoint paths from  $s$  to  $t$ . Since no red edge is contained in two of those, this corresponds to  $k$  internally disjoint paths from  $a$  to  $b$  in  $G$ : each path is of the form  $(s, v'_1, v''_1, \dots, v'_m, v''_m, t)$  and corresponds to  $(a, v_i, \dots, v_m, b)$  in  $G$ .

Now assume  $k_0 < k$  and consider an  $(s, t)$ -cut  $(S, T)$  of  $G'$  of size  $k_0$

Claim: We may assume that every edge in  $\delta^+(S) \setminus \{s, t\}$  is red:

If an edge entering  $v'_i$  is in  $\delta^+(S)$ , move  $v'_i$  from  $T$  to  $S$  without changing the size of the cut. Similarly if an edge leaving  $v''_i$  is in  $\delta^+(S)$ , move  $v''_i$  to  $T$ . If  $\{s, t\} \notin E(G)$ , then removing the  $k_0$  vertices of  $G$  that correspond to red edges in  $\delta^+(S)$  disconnects  $G$  (no path from  $s$  to  $t$  in  $G'$  after removal  $\implies$  none from  $a$  to  $b$  in  $G$ )  $\nmid$ .

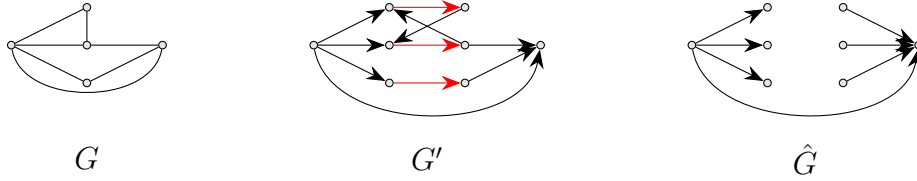
If  $\{s, t\} \in E(G)$ , proceed as follows:

Let  $A$  be the set of vertices in  $G$  corresponding to red edges in  $\delta^+(S)$  in  $G'$  ( $|A| = k_0 - 1$ ). Let  $\hat{G}$  be the graph obtained by removing  $A$ .

If  $\hat{G}$  not connected  $\nmid G$  is  $k$ -vertex connected.

Thus  $\hat{G}$  connected. Since  $G$  has  $\geq k + 1$  vertices either  $a$  or  $b$  has an additional neighbour.

Wlog  $a$  does  $\implies \hat{G} - a$  is not connected  $\nmid G$  is  $k$ -vertex connected (see graph(2)).



□

**Definition 4.13.** A graph is **bipartite** with parts  $A, B$  if  $V(G) = A \cup B$  and every edge in  $G$  has one end in  $A$  and one in  $B$ .

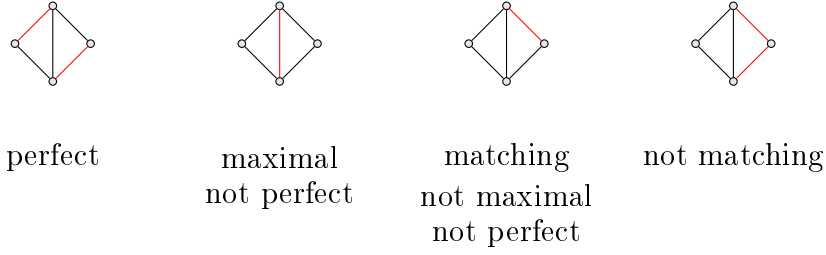
**Example 4.14.** •  $K_{a,b}$

•  $Q_d, d \geq 1$

**Definition 4.15.** A **matching** in a graph  $G$  is a set  $M \subset E(G)$  s.t. no two edges in  $M$  are incident.

A matching  $M$  is **perfect** if every vertex in  $G$  is incident to an edge in  $M$ .

**Example 4.16.**



**Theorem 4.17** (Hall's marriage theorem). *Let  $G = (V, E)$  be a bipartite graph with part  $A, B$ . Then  $G$  has a perfect matching iff  $\forall P \subset A, |N_G(P)| \geq |P|$  and  $\forall P \subset B, |N_G(P)| \geq |P|$ .*

*Proof.* " $\implies$ " Assume (Wlog)  $\exists P \subset A$  s.t.  $|N_G(P)| < |P|$ . Let  $M$  be a perfect matching. Then  $\forall v \in P$  the other endpoint of the matching edge incident to  $v$  is in  $N_G(v) \subset N_G(P)$ , and these vertices are distinct  $\implies |N_G(P)| \geq |P|$ .  
 " $\impliedby$ " Note that the condition implies  $|A| = |B|$ , since  $|B| \geq |N_G(A)| \geq |A|$  and vice versa. Define an auxiliary graph  $G' = (V', E')$  with:

- $V' = V \cup \{s, t\}$
- $E' = \{\{s, a\} : a \in A\} \cup \{\{b, t\} : b \in B\} \cup \{\{a, b\} : a \in A, b \in B\}$
- $\forall e \in E : c(e) = 1$

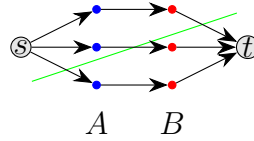
Every matching  $M$  in  $G$  defines an integral  $(s, t)$ -flow ( $f(e) \in \{0, 1\}$ ) in  $G'$  with  $\text{val}(f) = |M|$  and vice versa.

Let  $(S, T)$  be an  $(s, t)$ -cut. Let  $A_S = A \cap S, A_T = A \cap T, B_S = B \cap S, B_T = B \cap T$ . It follows  $|\delta^+(S)| \geq |A_T| + |B_S| + |N_{A_S}(B_T)| \geq |N_{A_T}(B_T)| + |B_S| + |N_{A_S}(B_T)| =$

$$|N_A(B_T)| + |B_S| \stackrel{\text{condition}}{\geq} |B_T| + |B_S| = |B| = |A|.$$

Thus by maxflow-mincut,  $\max_{f \text{ flow}} \text{val}(f) = \max_{f \text{ integral flow}} \text{val}(f) \geq |B|$ .

Therefore there exists a matching in  $G$  of size  $\geq |B|$ , i.e. a perfect matching.



□

## 5 Coloring

**Definition 5.1.** Let  $k \in \mathbb{N}$ . A (proper)  **$k$ -vertex-coloring** ( $k$ -coloring) of a graph  $G$  is a function  $c : V(G) \rightarrow \{1, 2, \dots, k\}$  such that for every edge  $\{u, v\} \in E(G)$ , we have  $c(u) \neq c(v)$ . The **chromatic number**  $\chi(G)$  of a graph  $G$  is the smallest integer  $k$  such that  $G$  has a proper  $k$ -coloring. We call  $G$  **colorable** if it has a  $k$ -vertex-coloring.

Write  $\Delta(G)$  for the maximum degree of a vertex in  $G$ .

**Lemma 5.2.** *Let  $G$  be a graph. Then  $\chi(G) \leq \Delta(G) + 1$ .*

*Proof.* Order the vertices of  $G$  as  $v_1, v_2, \dots, v_n$ . We will color the vertices in this order. When we come to color  $v_i$ , it has at most  $\Delta(G)$  neighbors, so there are at most  $\Delta(G)$  colors that are forbidden for  $v_i$ . Since we have  $\Delta(G) + 1$  colors available, there is always at least one color that we can use to color  $v_i$ .  $\square$

**Remark 5.3.** The bound  $\chi(G) \leq \Delta(G) + 1$  is tight with equality for exactly the complete graphs  $K_n$  and the odd cycles  $C_{2k+1}$ .

**Lemma 5.4.** *Let  $G$  be a (connected) graph. If there is  $v \in V(G)$  such that  $\deg_G(v) < \chi(G)$ , then  $G$  has a coloring with  $\Delta(G)$  colors.*

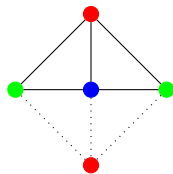
*Proof.* Proof by Induction on  $|V(G)|$ .

Base case: If  $|V(G)| = 1$ , then  $\chi(G) = 1$  and  $\Delta(G) = 0$ , so the statement holds.

Induction step: Assume the statement holds for all graphs with fewer than  $n$  vertices, and let  $G$  be a graph with  $n$  vertices. Let  $v \in V(G)$  such that  $\deg_G(v) < \chi(G)$ . Consider the graph  $G' = G - v$ . Since  $G'$  has fewer vertices than  $G$ , by the induction hypothesis, we have  $\chi(G') \leq \Delta(G')$ . Now let's look at two cases:

(1)  $\exists w \in V(G)$  such that  $\deg_G(w) = \Delta(G)$  and  $w$  is a neighbour of  $v$  in  $G$  and there is no  $w' \in V(G')$  with  $\deg_{G'}(w') = \Delta(G)$  not a neighbour of  $v$ . Then it follows that  $\Delta(G) = \Delta(G') + 1 = \chi(G') + 1$  and  $\chi(G) = \chi(G') + 1$ . Therefore  $\Delta(G) = \chi(G)$ .

(2) There is no  $w \in V(G)$  with  $\deg_G(w) = \Delta(G)$  s.t.  $w$  is a neighbour of  $v$  in  $G$ . Then  $\Delta(G) = \Delta(G')$  and  $\chi(G) = \chi(G')$ . Therefore  $\Delta(G) = \chi(G)$ .



$\square$

**Theorem 5.5** (Brooks' Theorem). *Let  $G$  be a connected graph. If  $G$  is neither an odd cycle nor a complete graph, then  $\chi(G) \leq \Delta(G)$ .*

*Proof.*  $\Delta(G) = 0$ : Then  $G$  is a single vertex and therefore complete.

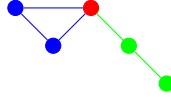
$\Delta(G) = 1$ : Then  $G$  is a path and therefore complete.

$\Delta(G) = 2$ : Then  $G$  is a cycle. If  $G$  is an odd cycle, then  $\chi(G) = 3 = \Delta(G) + 1$ . If  $G$  is an even cycle, then  $\chi(G) = 2 = \Delta(G)$ .

Let  $\Delta(G) \geq 3$

By Lemma ("5.4"), we may assume that  $G$  is  $\Delta(G)$ -regular ( $\forall v \in V(G), \deg_G(v) = \Delta(G)$ ).

- (i)  $\exists v \in V(G)$  such that  $G - v$  is disconnected. Then  $G - v$  has at least two components. Let  $C_1, \dots, C_k$  be the components of  $G - v$ , and let  $G_i = G[V(C_i) \cup \{v\}]$ .  $G_i$  is connected,  $\Delta(G_i) = \Delta(G)$  and since  $\deg_{G_i}(v) < \deg_G(v) = \Delta(G) \implies$  (by Lemma "5.4") we can color  $G_i$  with  $\Delta(G)$  colors. Fix such a coloring and then permute the colors s.t.  $v$  always has the same color. Then this is a coloring of  $G$  with  $\Delta(G)$  colors.



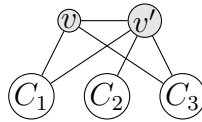
- (ii) Wlog  $G$  is connected.

Assume not (1) and  $\exists v, v' \in V(G)$  s.t.  $G - v - v'$  is disconnected.

Let  $C'_1, \dots, C'_k$  be the components of  $G - v - v'$ . Let  $G_i = G[V(C'_i) \cup \{v, v'\}]$ . Let  $G'_i$  be the graph obtained from  $G_i$  by adding  $\{v, v'\}$  if it is not already there.

Observe that  $\forall i \in \{1, \dots, k\}$ ,  $v$  has a neighbour in  $C_i$  (say  $v'_i$ ).

Case 1:  $k \geq 3$ :  $G'_i$  is connected and  $(*) \deg_{G'_i}(v) \leq \deg_G(v) + 1 - 2 = \deg_G(v) - 1$  (since  $v$  has at least two neighbours in  $G_i$ ). We can color  $G'_i$  with  $\Delta(G)$  colors (by Lemma "5.4"). Fix such a coloring and then permute the colors s.t.  $v, v'$  always have the same color. Then this is a coloring of  $G$  with  $\Delta(G)$  colors.



Case 2:  $k = 2$ : Unless  $\{v, v'\} \in E(G)$  and both  $v, v'$  have degree precisely 1 in  $G_1$  or  $G_2$ , we can argue as in Case 1. Now Wlog  $\deg_{G_1}(v) = \deg_{G_1}(v') = 1$ . Let  $N_{G_1}(v') = \{v''\}$ . Observe that  $G - v' - v''$  is not connected. Pending...

□

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